

Name: _____

Date: _____

Badge 1 Boolean Logic Worksheet

This worksheet is required as part of the Badge 1 Curriculum for FRC Team 4451. Complete this to the best of your ability, and return to mentors for review.

1. Statements and Predicates

Write **proposition**, **predicate**, or **instruction** beside each item. Elaborate on your choices.

Item	Classification
The robot is enabled.	_____
<code>temperature < limit</code>	_____
Set the launcher voltage.	_____

For `temperature < limit`, identify both values you need before deciding whether it is true or false.

Answer:

2. Operators and Truth Tables

Fill every blank with **true** or **false**. Here, p and q stand for propositions. Keep the rows in the order shown.

Symbol	Read it as
!	NOT
& &	AND
	OR (one or both)
^	XOR (exactly one)
→	If ... then ...
↔	If and only if

p	q	$!p$	$p \& \& q$	$p \mid \mid q$	$p \wedge q$
true	true	_____	_____	_____	_____
true	false	_____	_____	_____	_____
false	true	_____	_____	_____	_____
false	false	_____	_____	_____	_____

p	q	$p \rightarrow q$	$p \leftrightarrow q$
true	true	_____	_____
true	false	_____	_____
false	true	_____	_____
false	false	_____	_____

3. Equivalent Expressions

Rewrite the first two expressions using De Morgan's laws so that $!$ applies to individual names.

Rewrite $p \rightarrow q$ using only $!$ and $||$.

Expression	Equivalent expression
$!(\text{enabled} \ \&\& \ \text{connected})$	_____
$!(\text{fault} \ \ \text{disabled})$	_____
$p \rightarrow q$	_____

Classify $p \ || \ !p$ as a **tautology** or **contradiction**: _____

Classify $p \ \&\& \ !p$ as a **tautology** or **contradiction**: _____

4. Implication and its Relatives

For this exercise, assume this statement is true: **If a student earns full credit, then the student submitted the worksheet.**

Form	Write the statement	Equivalent to the original?
Converse	_____ _____	_____
Inverse	_____ _____	_____
Contrapositive	_____ _____	_____

5. Rules of Inference

Treat the given statements as **TRUE**. Write the conclusion and name the rule. You may look up rule names in the lesson; explain why the conclusion follows. **Modus Ponnens and Modus Tollens are required.**

A. If a student earns full credit, the student submitted the worksheet. Morgan earned full credit.

Conclusion: _____

Rule: _____

B. If a student earns full credit, the student submitted the worksheet. Jordan did not submit the worksheet.

Conclusion: _____

Rule: _____

C. If a student completes the project, the student submits a repository. If the student submits a repository, the mentor can review the code.

Conclusion: _____

Rule: _____

D. The fault is in the cable or the sensor. The cable is not faulty.

Conclusion: _____

Rule: _____